

Diversity and Cyber Security Expertise: Policy report

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Executive summary

Our society is deeply and increasingly reliant on cyber security practitioners to ensure that digital services are dependable, critical infrastructure is resilient and sensitive data is protected. Their success depends on diverse ways of thinking, diverse skillsets, perspectives and backgrounds. Today, the profession is being transformed, with standardised categories of specialism, new certifications and evolving governance structures. While this process of professionalisation is driven by the need to better support practitioners and their employers, it is vital to understand any unintended consequences that may be emerging. We focus here on potential negative impacts on diversity, a topic that must be carefully addressed to ensure that cyber security remains an attractive field to work in, to support the legitimacy of professional institutions and to enhance the efficacy of the field.

We developed methodologies for analysing the alignment of practitioners with new classifications of cyber security specialisms, and for analysing the alignment of specialisms with cyber security problems. We tested these approaches using data from a small exploratory survey. We found that there is profound uncertainty about the impacts of professionalisation on diversity. This is rooted in a lack of data and a lack of theoretical analysis. This report takes a small step to correct this issue.

Recommendations

Surveys on diversity in cyber security (Decrypting Diversity) should be revived to ensure good quality data is available. They should be extended to gather more detailed data on respondents' expertise, for example using the CyBOK (Cybersecurity Body of Knowledge) framework¹. This would enable evidence to be gathered about correlations between diversity characteristics and the alignment of professionals with the UK Cyber Security Council Cyber Careers Framework specialisms. Regular surveys would enable the analysis of how this alignment is changing over time.

The UK Cyber Security Council should commence work on new candidate specialisms to address the current lack of coverage of security human factors and security awareness. The latter could be based on the European Cybersecurity Skills Framework 'Cybersecurity educator' role².

¹<https://www.cybok.org/>

²<https://www.enisa.europa.eu/publications/european-cybersecurity-skills-framework-ecsf>

Future iterations of CyBOK should be informed by research into expertise diversity, to ensure that the value of the humanities and social sciences within cyber security is better reflected. Humanities and social science researchers in the cyber security field should engage more closely with the CyBOK community to support this.

The UK Cyber Security Council's Cyber Career Framework's cross-reference of specialisms with CyBOK knowledge areas should be refined and validated through qualitative studies of cyber security problem scenarios.

About

This Cyber Expertise Diversity Survey was run by the Centre for Interdisciplinary Methodologies (CIM) in collaboration with the Research Institute for Sociotechnical Cyber Security (RISCS), led by Matt Spencer, RISCS Senior Fellow and Associate Professor at CIM.

Other CIM staff involved in putting the survey together and analysing the data include Carlos Cámara-Menoyo and Timothy Monteath.

The survey aims to inform cyber policy discussions by generating new insights into the value and distribution of cyber expertise.

Citing

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Chapter 1

Introduction

1.1 Background on this survey, aims and objectives

The Cyber Expertise Diversity Survey was designed to investigate how new structures being introduced into the cyber security profession may be influencing its composition. The survey is small scale, opportunistic, and exploratory. These limitations mean it is oriented more towards formulating and testing the questions we should be asking rather than providing conclusive answers. It aims to provide recommendations for the design of future larger studies.

The survey was conducted in association with RISCs (the Research Institute for Sociotechnical Cyber Security), and the approach was developed with help from a number of individuals in industry and government who supported the project by providing advice on the approach and objectives.

The Cyber Expertise Diversity Survey explores the intersection of diversity and professionalisation, and thus builds on a line of inquiry developed in recent years through the 2020-2021 NCSC/KPMG ‘Decrypting Diversity’ surveys of individual professionals, and other work such as the organisation-level ‘Cyber Security Skills in the UK Labour Market’ survey, which introduced additional diversity-focused analysis in 2021¹.

1.2 Research Goals

We wanted to understand how well aligned categories of cyber specialism are with EDI goals of the profession. We wanted to think through who will find it harder or easier to attain recognised status, and to examine how professionalisation is likely to change the distribution of cyber security expertise.

¹See <https://www.ncsc.gov.uk/report/diversity-and-inclusion-in-cyber-security-report>, <https://www.ncsc.gov.uk/report/decrypting-diversity-2021-diversity-and-inclusion-in-cyber-security>, <https://www.gov.uk/government/publications/cyber-security-skills-in-the-uk-labour-market-2024/cyber-security-skills-in-the-uk-labour-market-2024-technical-report>

Motivating these questions is a set of concerns that we believe require further evidence and analysis.

Firstly, scholars have observed that there is a tension between two goals of professional bodies like the UK Cyber Security Council (UKCSC). On the one hand, such bodies seek to standardise professional categories, statuses and processes of recognition. On the other hand, they have a moral purpose to ensure that opportunities are not unequally distributed across society. Standardising professional categories, statuses and processes of recognition can serve to entrench existing inequalities and create new forms of inequality (Evetts 2012; Daniel 2007). The effectiveness of professional bodies' efforts to support equality, diversity and inclusion (EDI) in their domain is crucial for their legitimacy and sustainability, and the success of these efforts will depend on having a good understanding of the actual and potential effects of standardisation.

Our second concern arose in conversations with experts in security awareness and human factors, and relates to the specific way in which expertise in people and culture is treated in the UKCSC's Cyber Careers Framework. In this framework, expertise in people and culture is treated as a generic requirement ('soft skills'); while this decision has the virtue of emphasising that all cyber security professionals need some forms of competence in this area, it fails to recognise expertise in people and culture as an area of deep expertise in its own right². Such expertise in people and culture does not currently lead to accreditation: anecdotally (and our survey was not of a scale that could substantiate this), there is a concern that women tend to be over-represented in this area of the profession (for instance, working in security awareness and human factors) compared with in other areas; if this is the case, the Cyber Careers Framework is at risk of exacerbating issues of gender representation by making it harder for women to gain recognition.

Finally, we note that many of the senior leaders in the profession came into cyber security before most of the categories of expertise and specialism that are now in common use were clearly defined, before degrees and other qualifications in cyber security existed, and indeed before the term 'cyber security' was widely used. Their career routes were often highly idiosyncratic. The UKCSC was founded to provide more structure, with the goal of better supporting the workforce. An early version of the UKCSC Cyber Careers Framework featured a 'Cyber Security Generalist' specialism, but this was removed in later iterations. Idiosyncratic career moves may be disincentivised by a structure that ties recognition to advancement in specific domains. Better understanding diversity of expertise may help us to understand the value of atypical expertise profiles, and the risks associated with incentivising more homogeneous skillsets. This was a further motivation for examining the methodologies and data that may be needed to understand the implications of professionalisation.

²We point the reader to the breadth of research that has taken place at RISCs, for an appreciation of the deeply technical nature of research into people and culture and related sociotechnical aspects of cyber security. See <https://riscs.org.uk/publications/>

1.2.1 Approach

Because the survey is exploratory, this report presents the data alongside methodological reflections and reflections drawing on existing social scientific literature. We examine:

- What kind of analysis would illuminate the effects of professionalisation on social inequalities?
- What kind of analysis would illuminate the value of diversity to the profession?
- Are there indications in our dataset that would support specific concerns about the impact of professionalisation on diversity?

Chapter 2

The Cyber Expertise Diversity Survey

2.1 Design and distribution

To address our research aims, we designed a survey to gather data on the diversity of expertise within the cyber security profession, in the UK, and on the value of diverse expertise in relation to the problems cyber security practitioners face in their work. The *Cyber Expertise Diversity Survey* consisted of a total of 21 questions grouped in the following blocks:

1. A first block about the respondent, where we asked them 5 questions about their job and workplace (i.e. role, number of years of experience, size and sector of their workplace) followed by an optional set of six questions about EDI (i.e. age bracket, disabilities, ethnic origin, gender and sexual orientation).
2. A block called “Expertise profile” (or expertise comb -see Section 2.1.1 below), where we asked participants to rate their perceived expertise around 26 areas, 21 areas identified by the Cyber Security Body Of Knowledge (CyBOK) (Martin et al. 2021) and 5 extra categories added by the authors¹.
3. A block called “Problems or Challenges” (or scenarios comb), where we asked participants to describe one or more scenarios from their working experience in which resolving a problem required a combination of forms of expertise to be brought together. For every scenario, respondents were asked to provide a brief title and description, state whether it was related

¹The 26 categories used were: adversarial behaviours, applied cryptography, authentication authorisation accountability, cloud security, cryptography, cyber physical systems, cyber security law regulation, data science ai for cyber security, distributed systems security, forensics, formal methods for security, hardware security, human factors and usable security, malware attack technologies, network security, operating systems virtualisation security, physical layer and telecommunications security, political aspects of cyber security, privacy online rights, risk management governance, secure software lifecycle, security culture narrative and communication, security operations incident management, social organisation strategy and process, software security, web mobile security

to their current job (and if negative, to provide information about job and workplace), and rate the amount of expertise needed to address the scenario, using the same 26 categories used in the previous block.

4. An optional block with free text comments, where we asked participants to list any areas of expertise that they may consider to be missing, and to provide any feedback about the survey.

The survey was created using Qualtrics, and was configured to only accept responses from people based in the UK, and not to collect personal details. Ethical approval was given by the University of Warwick’s Humanities and Social Sciences Research Ethics Committee (HSSREC)². To ensure informed consent, we designed the survey to include key information for participants about purpose and data use in the initial pages, as well as a separate downloadable participant information leaflet. Because we did not collect personal identifying data in the survey, we generated a random identifier for participants to store, should they wish to withdraw from the study or get in touch with us about their responses at a later date. Participants were not offered remuneration for their time.

The survey was accepting responses between 12/09/2024 and 19/11/2024. Because this was a resource-constrained exploratory project, we followed a convenience sampling approach, distributing the survey via cyber security industry mailing lists and via the LinkedIn social media network (see Section 2.3.1 below).

Anonymised results can be found in a companion repository (Cámara-Menoyo, Spencer, and Monteath 2025). The repository contains additional material such as metadata describing every variable, the scripts used to clean the data, and a PDF version of the survey, following FAIR principles (Wilkinson et al. 2016) and released under an open licence, to foster discoverability and reuse.

2.1.1 The comb concept

The survey is built around the metaphor of a ‘broken comb’ of expertise, with each ‘tooth’ in the comb representing one area of expertise. For blocks 2 and 3, we asked survey participants to imagine their expertise as a kind of comb where each ‘tooth’ of the comb represents one area of specialism. Some cyber security professionals have one or two areas in which they have a lot of expertise, other areas in which they have a little, and some in which they have none. Some people have a broad base; others are highly specialised. It can depend on domain, career stage, and individual personality.

²<https://warwick.ac.uk/services/ris/research-integrity>

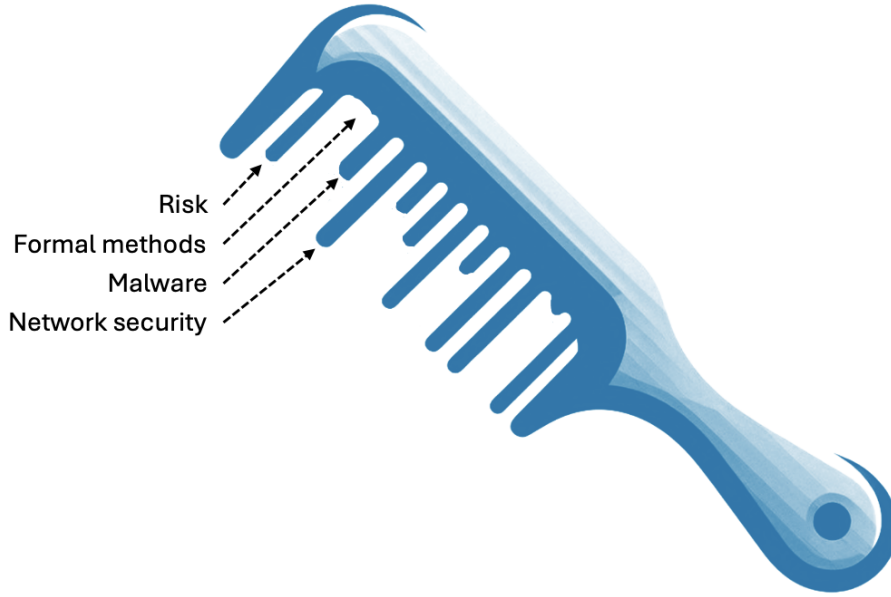


Figure 2.1: The broken comb metaphor, which was suggested to us by John R.

The survey used an interface based on a series of sliders through which participants could express how they saw their own ‘broken combs’ of expertise, and to define similar ‘broken combs’ reflecting the combinations of expertise required to address cyber security problems and challenges that they tackle in their work.

We based the ‘teeth’ of the comb on the knowledge area classification developed by the ‘CyBOK’ project (Martin et al. 2021), extended slightly to account for areas that are not accounted for in the standard CyBOK classification (see Section 4.2.1).

We thus gathered three types of combs to support the analysis.

- The **personal comb** is just the values that a person put in for their own expertise using the sliders. For alignment calculations we only use the CyBOK values.
- The **specialism comb** is a representation of the Cyber Careers Framework’s CyBOK mapping: we create these by assigning a numeric value to what the UKCSC considers to be ‘core’, ‘related’ or ‘wider’ relevance to each specialism.
- The **scenario comb** is the values that a person put in for that specific scenario using the sliders. For alignment calculations we only use the CyBOK values.

2.1.2 The alignment metric calculation

In order to see how well (or badly) are two combs aligned, we compute an **alignment index** $A_{(x,y)}$ as follows.

Given a set of **test values** $X : [x_0, x_1, \dots, x_n]$ and a corresponding set of **reference values** $Y : [y_0, y_1, \dots, y_n]$, the alignment index will be:

$$A_{(x,y)} = \frac{\sum I(X)}{\sum Y}$$

Where the indicator function $I(X)$ is defined as follows:

- $I(X) = x_m$ when $x_m \leq y_m$
- $I(X) = y_m$ when $x_m > y_m$

The indicator function is used to prevent a test comb getting ‘credit’ for overshooting the reference values (which could otherwise misleadingly offset deficits in other areas). To do so, it takes the reference value as a ceiling in each tooth of the test comb. This also means that the maximum possible alignment is 1.

With this, we will be calculating two types of alignments:

- **Personal alignment**, where test values (X) are the personal comb (responses) and the reference values (Y) are the specialism comb. This gives us a sense of how close respondents’ combs are to the different kinds of specialism.
- **Scenario alignment**, where test values (X) are the specialism comb and the reference values (Y) are the scenario comb. In this case, we are interested in how well the specialisms as defined in the Cyber Careers Framework provide the kind of expertise required to address each problem defined in the provided scenarios. For instance, if a problem scenario is defined by a respondent to require only expertise in Adversarial Behaviour and Forensics, the Cyber Security Management specialism would be best aligned of all the UKCSC specialisms, as this is the only specialism for which both of these knowledge areas are defined as ‘core’ areas of expertise. We look at this for individual specialisms, and also look at the alignment of combinations of specialisms in permutations up to 5.

In both cases, the **alignment index** will result in a number between 0 and 1, where 1 means perfect alignment between the test value and the reference value and 0 means not aligned at all.

Computations for the
alignment metric can be
audited at the file
`scripts/alignment_metric.R`

2.2 Results

The survey was conducted between 12/09/2024 and 19/11/2024 and received responses from 56 individuals, describing 65 scenarios.

In this section we will be providing an overview of the responses according to each of the blocks of the survey. Detailed analysis and interpretations of those results can be found in the following sections of the report and the appendices.

2.2.1 Respondents’ demographics

In terms of demographics, we can see (Table 2.1) that, more than 50% of the responses are between 40 and 59 years old.

Table 2.1: Responses by age

Age	N	Percent
20-25	4	7.1%
26-29	1	1.8%
30-39	10	17.9%
40-49	15	26.8%
50-59	18	32.1%
60 and over	5	8.9%
I prefer not to answer	3	5.4%

Table 2.2 and Table 2.4 shows a clear preponderance of male respondents over other genders. Women account for a 17.9% of respondents to our survey - which is inline with the workforce estimates of the UK Cyber Skills Survey (17%) (DSIT 2025).

Table 2.2: Responses by gender

Gender	N	Percent
Male	41	73.2%
Female	10	17.9%
I prefer not to answer	4	7.1%
Non binary	1	1.8%

Table 2.3 brings together responses by age and gender in a two way table.

Table 2.3: Two way table of responses by gender and age

	Men	Women and Non-binary
20-25	3	1
26-29	1	0
30-39	5	5
40-49	10	4
50-59	16	1
60 and over	5	0
I prefer not to answer	1	0

Table 2.4: Responses by gender (2)

Trans	N	Percent
No	51	91.1%
I prefer not to answer	4	7.1%
Yes	1	1.8%

In terms of ethnicity (Table 2.5), we can see a clear prevalence of White respondents (76.8%), followed by far by Asian and mixed race. Compared to the estimates of the UK Cyber Skills Survey our survey has slightly less White respondents overall (81%) (DSIT 2025)⁸¹

Table 2.5: Responses by ethnicity

Ethnicity	N	Per-cent
White	43	76.8%
Asian (Indian, Pakistani, Bangladeshi, Chinese, any other Asian background)	4	7.1%
Mixed two or more ethnic groups	3	5.4%
Prefer not to say	3	5.4%
Black/African/Caribbean	1	1.8%
Other (Arab or any others)	1	1.8%
NA	1	1.8%

In terms of disabilities (Table 2.6), respondents reporting some type of disability account for a total of 17.8%. Compared to the UK Cyber Skills Survey this is a slightly above the estimates for people with a disability as part of the workforce (8%) but is inline with figures for neuro-divergence (16%) which our survey did not specifically ask about (DSIT 2025).

Table 2.6: Responses by disabilities

Disabilites	N	Per-cent
No, I do not have a disability, or a history/record of having one	45	80.4%
I prefer not to answer	6	10.7%
Yes, I have a mental disability or neurodiversity, or have a history/record of having one	4	7.1%
Yes, I have both a mental and physical disability, or have a history/record of having them	1	1.8%

While the small sample and opportunistic sampling approach prohibit broader conclusions, we observe that these figures are broadly aligned with the sector (National Cyber Security Centre 2021; DSIT 2025). We will argue, however, that the data is suggestive that further data collection would be of value.

2.2.1.1 Job-related questions

In terms of job-related questions, we can see (Table 2.7) that half of our respondents have less than 10 years of experience, despite most of responses being from relatively mature respondees (see Table 2.1), and 71% have up to 20 years. 12.5% had more than 25 years of experience. This is in line with the relatively high age of the respondents, but may also suggest that some

respondents may have shifted their career towards cyber security less than 10 years ago.

Table 2.7: Responses by years of experience

Age	N	Percent
0-5	14	25.0%
6-10	14	25.0%
11-15	12	21.4%
16-20	5	8.9%
21-25	4	7.1%
25+	7	12.5%

Table 2.8: Responses by organisation size

Organisation Size	N	Percent
Large (250 employees or more)	40	71.4%
Medium (50 to 249 employees)	5	8.9%
Small (10 to 49 employees)	5	8.9%
Micro (1 to 9 employees)	6	10.7%

Table 2.9 brings together the two previous responses consider the alignment between years spent in the cyber security industry and the size of company at which respondents work.

Table 2.9: Two way table of responses by organisation size and years experience

	0-5	6-10	11-15	16-20	21-25	25+
Large (250 employees or more)	8	11	10	5	1	5
Medium (50 to 249 employees)	2	1	1	0	1	0
Micro (1 to 9 employees)	1	2	1	0	2	0
Small (10 to 49 employees)	3	0	0	0	0	2

Table 2.10: Responses by Industry sector

Industry Sector	N	Percent
Professional, Scientific, and Technical	14	25%
Information and Communications	9	16.1%
Utilities and Production (Including Manufacturing)	9	16.1%
Other (low n)	8	14.3%
Other (Please State)	7	12.5%
Education	5	8.9%
Transport and Storage	4	7.1%

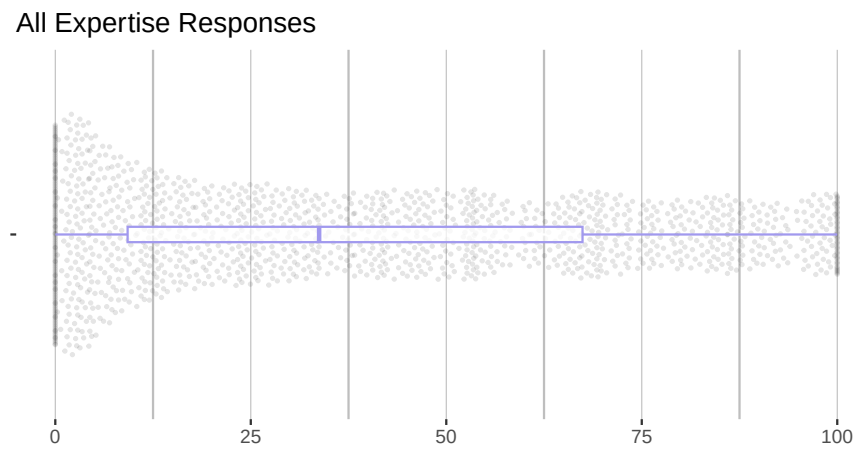


Figure 2.2: Participants' responses on their personal expertise combs, in aggregate

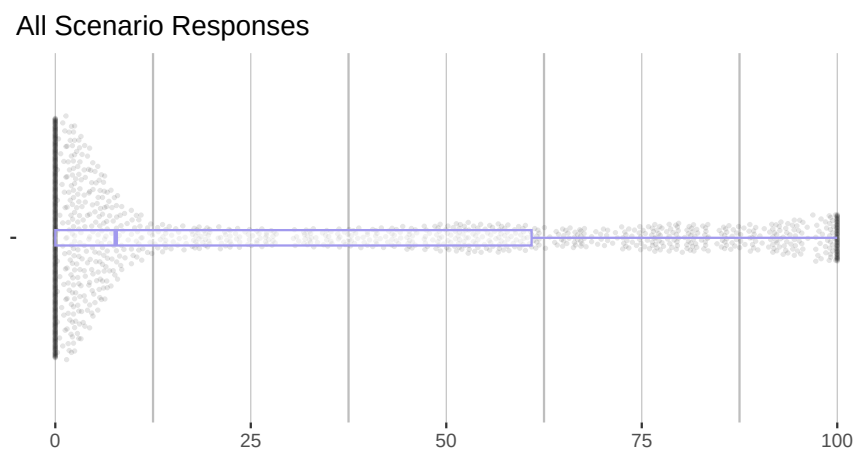


Figure 2.3: Participants' responses on their personal scenario combs, in aggregate

2.2.2 Comb responses

The visualisations above are beeswarm plots overlaid with a box and whisker plots showing all the responses in aggregate. For a more detailed view, refer to Appendix A and Appendix B.

Figure 2.2 shows all the personal comb responses. Respondents were using the full range of the sliders, with 10.1% responses being set to 0 and 3.4% set to 1. Figure 2.3 shows that respondents used the full range of the sliders to specify the needs of scenarios, but tended to leave more sliders at a low value when compared with personal combs.

Figure 2.4 below shows how the personal score for each of the 26 categories was distributed. Breaking down the responses by expertise category, we observe that our additional 5 non-CyBOK categories featured in respondents' profiles as commonly as CyBOK categories did. This helps to validate that they were relevant and recognisable.

Figure 2.5 shows that when describing scenarios, respondents tended to use a subset of the 'teeth' and the other sliders at a low value. The scenarios elicited from respondents favoured areas such as *security culture*, *adversarial behaviours*, *risk management* and *security operations*. This reflects the prominence of organisational challenges and incidents in the dataset.

2.3 Key limitations

Given resource constraints (we received no direct funding for this research), the project was conceived as an exploratory exercise, aiming to develop analytical tools suitable for investigating the value of diverse expertise in cyber security in the UK. It is important to bear in mind the following key limitations when interpreting the results.

2.3.1 Sampling

The total number of responses is modest (56). To put this into context, the government supported Decrypting Diversity 2020 report was based on 1252 responses, and the 2021 follow-up drew on a sample of 945.

Our sampling strategy was opportunistic: we distributed the invitation to participate via a number of cyber security related industry networks within the UK and on LinkedIn. A larger, random-sampling approach would have many benefits but would require much more significant investment. A convenience approach was chosen as appropriate for exploratory analysis demonstrating the value of an analytical frame that brings questions about the distribution and value of diverse expertise into conversation with questions about diversity of personal characteristics. Small sample studies like this should be seen as part of the wider endeavour to understand the domain, as opportunities for lightweight prototyping of methodologies for data collection and analysis.

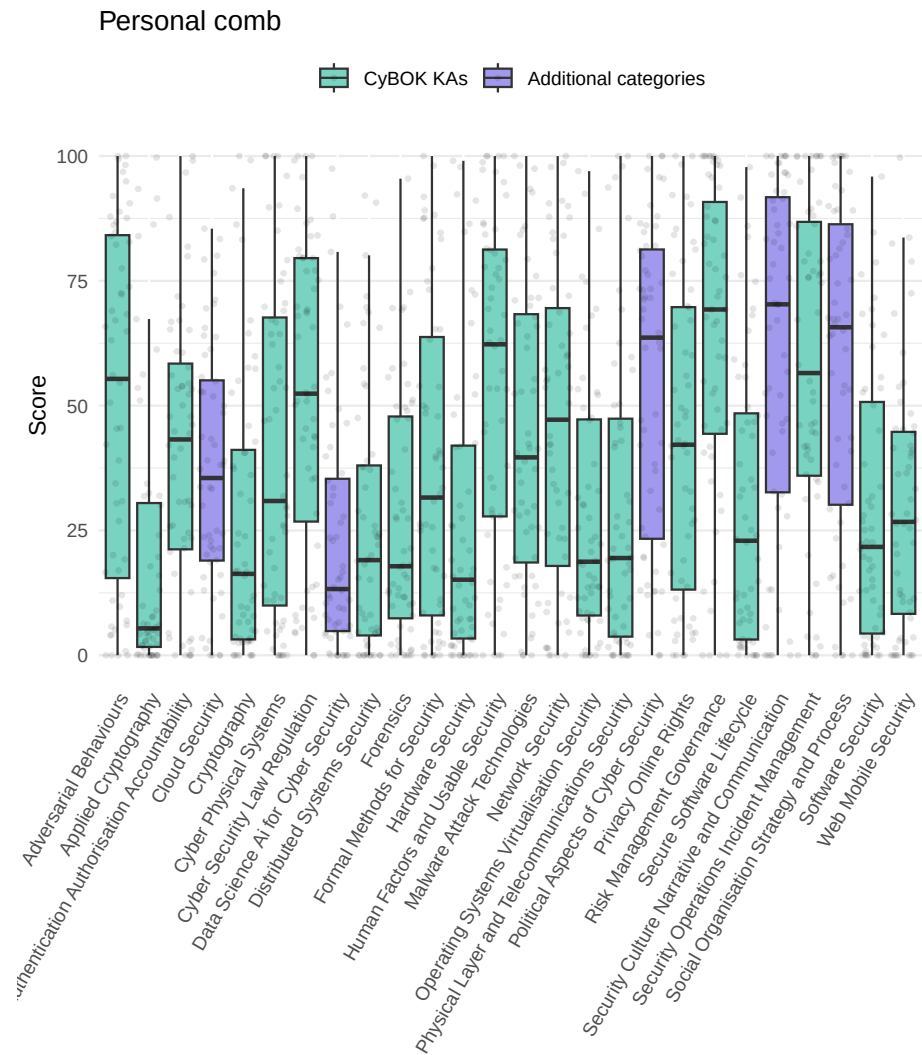


Figure 2.4: Overall distribution of personal comb scores (blue: CyBOK KAs; green: additional categories)

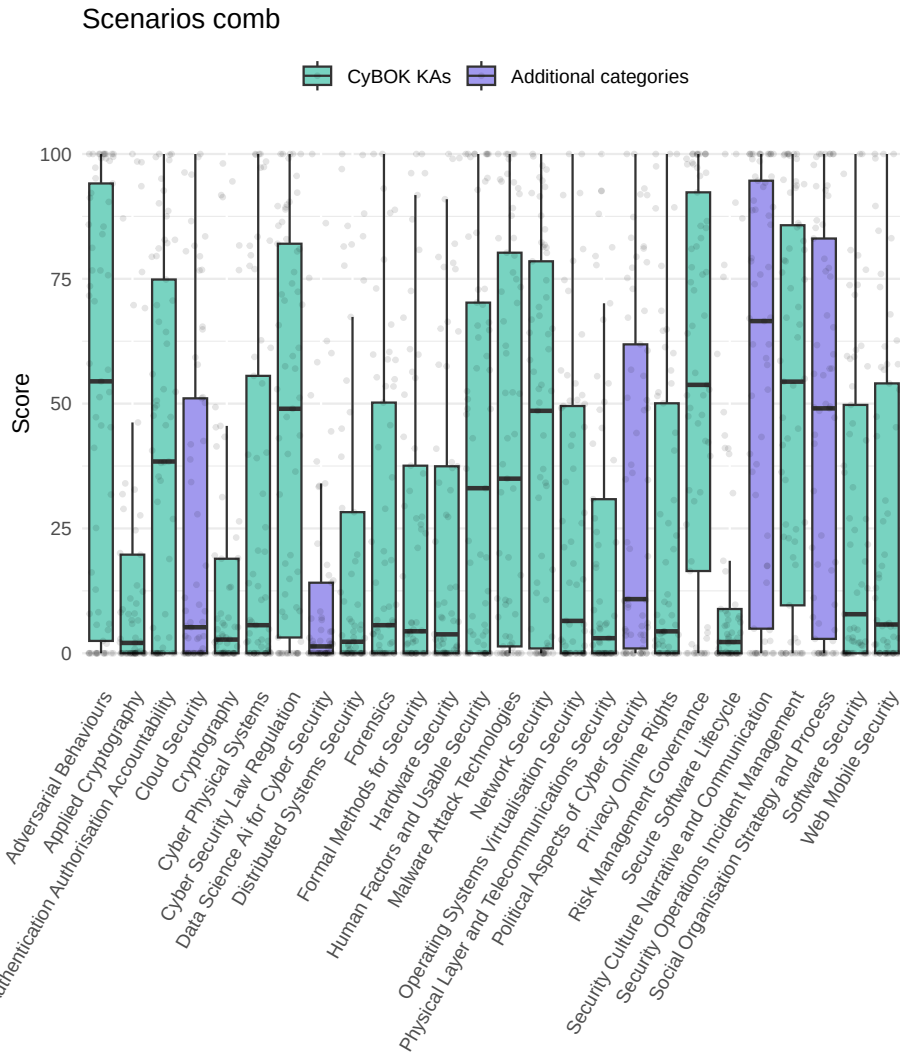


Figure 2.5: Overall distribution of scenarios comb scores by category

2.3.2 Sliders

Given the challenges in establishing an absolute scale across all ‘teeth’, we asked participants to focus on the relative proportions of each category, the focus thus being on the ‘shape’ of expertise rather than its absolute extent. Because sliders generate numerical values, they allow for analysis of these shapes, at individual or aggregate levels, but it must be remembered that the scales being applied are inevitably imprecise.

A further limitation of the use of relative slider was that their use by respondents created a non-parametric distribution to the data collected. As the beeswamp plots Figure 2.2 and Figure 2.4 showed, while respondents used the whole range of each slider available, on aggregate, responses clustered around the tails of each distribution and at median and quartile points. Consequently, the overall distribution of responses has ‘three’ humps. The non-standard distribution of this data, along with our small sample size, made the application of traditional statistical methods - for which a standard distribution is assumed - unsuited to our data. Furthermore, the way in which the use of the sliders amongst respondents from different categories, such as by gender, differed further significantly complicating statistical analysis between these groups.

Figure 2.6 shows the ‘three’ hump pattern of the data and the differing density of responses by gender. We consider the miss-alignment between the two different groups to be a finding of how each group self-reports. Rather than seeking to ‘correct’ this data for statistical analysis we incorporate this into our ‘interpretive’ analysis of the responses to our survey.

Because CyBOK is designed to refer to established areas of cyber security knowledge, we assumed participants would be familiar with the categories, and able to provide a relative assessment of their own expertise. However, respondents will vary in their level of understanding of the CyBOK categories, and this will also introduce imprecision into the analysis. A better resourced survey might assess familiarity with the categories as part of the survey process, or single out a subset of respondents for an interview to examine levels of understanding.

2.3.3 Alignment scores

The data has some important limitations that should be kept in mind when interpreting alignment scores:

- The specialisms are defined according to ‘core’, ‘related’ and ‘wider’ relevance of categories of expertise, which to be used in our calculations have to be translated into positions in the slider. We have chosen values of 90, 60 and 30 as parameters for these levels of relevance respectively. Alternative parameters can be readily substituted in the calculation.
- The specialisms are defined according to the standard CyBOK knowledge area classification. The relevance of other forms of expertise is left undefined. This means that our additional teeth of the comb, representing areas that are not accounted for in CyBOK, must be ignored for the purposes of the alignment calculations. In certain cases, for instance where a scenario required significant amounts of expertise in one or more of the non-CyBOK areas, the alignment calculation could be misleading. We have highlighted

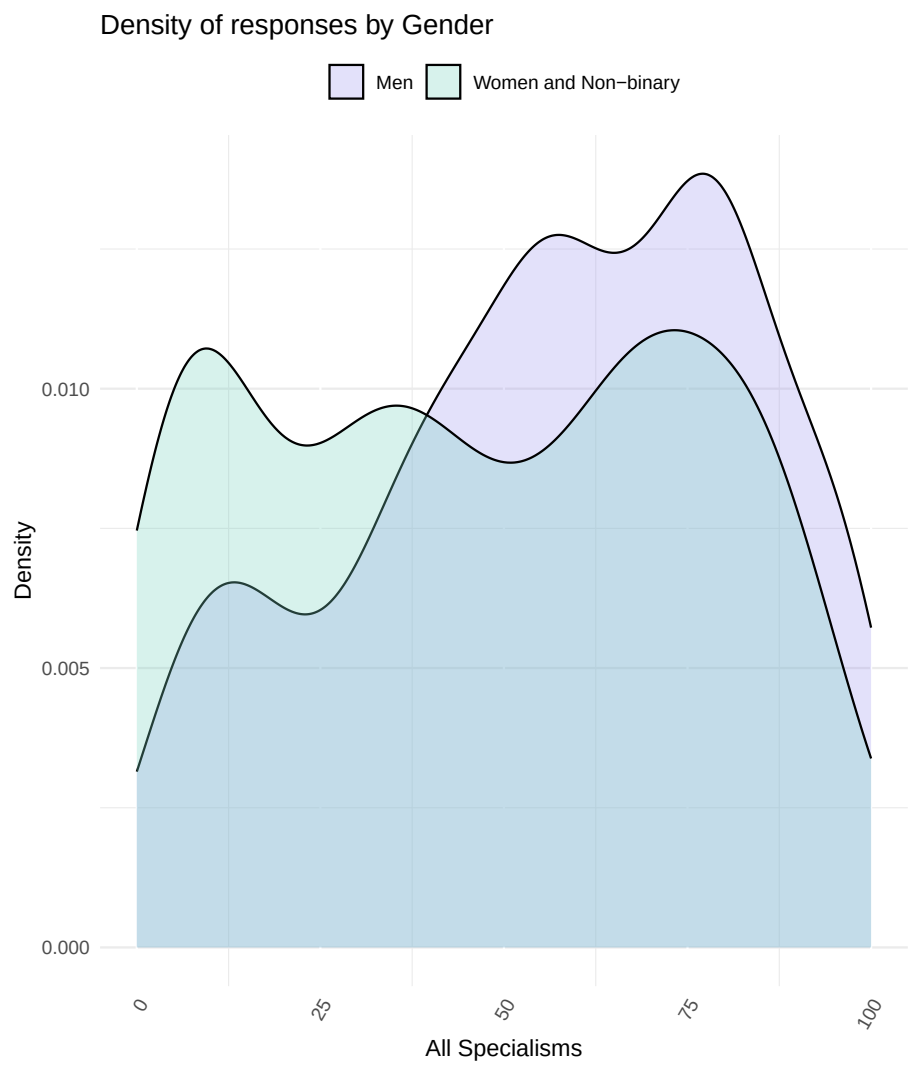


Figure 2.6: Density of responses by Gender

this in the analysis, and reflect on how reliance on established classifications like CyBOK creates challenges where one is interested in accounting for neglected or emerging areas of expertise.

Chapter 3

Professionalisation and Inclusion

3.1 Professionalizing Cyber Security in the UK

Concerns about the adequacy of the UK's cyber security workforce, and its ability to protect society, state and economy, have driven a number of interconnected policy efforts, aiming to:

- formalize cyber security education,
- bring more people into the field,
- give recognition to those with cyber security expertise,
- to ensure that adequate cyber security knowledge is integrated into related professions, and
- to enable employers to better understand what they need from their cyber security employees.

The first UK National Cyber Security Strategy, published in 2016, identified the need for government to do more to support the development of the cyber security profession¹. As part of the implementation of the Strategy, the Department for Digital, Culture, Media and Sport launched a 2018 proposal and consultation on the creation of the UK Cyber Security Council (UKCSC). The consultation recognised that there were existing forms of recognition, such as offered by the Institute of Information Security Professionals, which had received a Royal Charter in 2018, and the International Information System Security Certification Consortium (ISC2), which governs the international CISSP (Certified Information Systems Security Professional) qualification, but argued that there were outstanding issues with coverage (some kinds of cyber security expertise were not covered, or partially covered), and equivalence. The UKCSC was thus set

¹<https://www.gov.uk/government/publications/national-cyber-security-strategy-2016-to-2021>

up to “act as a governing voice for the cyber security profession in the UK” (Ministry of Defence 2021), and was launched in 2021.

Pillar 5 of the UK National Cyber Security Strategy 2022-2030, is the objective to ‘develop the right cyber security skills, knowledge and culture’ so that ‘sufficient, skilled and knowledgeable professionals fulfil all required cyber security needs’ (Cabinet Office 2022). This is to be delivered through “consistent taxonomies, standards and pathways” providing “diverse and fulfilling cyber security careers” and wider status recognition through accreditation and standardisation. The UKCSC’s work has thus become central to the national strategic vision.

In pursuit of these goals, the UKCSC developed a ‘Cyber Careers Framework’ which identifies four levels of expertise (Associate, Practitioner, Principal, and Chartered) across a set of 15 specialisms. Assessments of applicants for recognition are carried out by member organisations of UKCSC, who receive a license for specific specialisms: currently the Chartered Institute of Information Security, The Cyber Scheme, and CREST. Recognition is being delivered in a phased manner, and the first chartered cyber security professionals (ChCSP) were recognised in October 2023. The roll-out process has met with some difficulties, with only 6 specialisms currently available, and a relatively modest uptake so far. One challenge has been in the definition of specialisms, as some of these were inherited from pre-existing schemes such as the NCSC’s Certified Cyber Professional², whereas others have been defined from scratch. This has led to some inconsistencies in the style and extent of requirements across specialisms. It is widely recognised that the Cyber Careers Framework will need to evolve and improve over time, something that is necessary in any case to keep pace with changing technologies, threats and practices. Gathering and interpreting data will be important in enabling a responsive approach to cyber security problems and practices, on its workforce and the distribution of expertise, in order to maintain its alignment and relevance in the years ahead.

3.2 Social closure

Professionalisation can be analysed sociologically through the lens of social openness and social closure. The concept of social relationships being ‘open’ or ‘closed’ has deep roots in sociology, and is a building block of Max Weber’s classic analysis of social action (Weber 2019[1921]). Social closure is an umbrella term for the ways in which groups construct and maintain monopolies over opportunities, by including some people and excluding others. Understanding how and why closure occurs is key to our understanding of social structure and power. It can operate at multiple levels. Even where opportunities are formally open, groups may nevertheless be able to maintain control over resources through influencing access to jobs or status, for instance through cronyism or nepotism. Closure can create durable barriers to entry or ‘glass ceilings’ that limit opportunities. Social closure can be durable because, once they exist, monopolies on resources tend to create an alignment of interest among group members, as they have a common interest in maintaining the status quo and resisting change. Weber considered guilds to be an ‘archetype’ of closure where it is motivated by a cooperative monopolisation of (resource) acquisition, and

²<https://www.ncsc.gov.uk/news/forthcoming-closure-of-ccp-scheme>

under the rubric of ‘occupational closure’ the sociology of professions has studied professionalisation as an important example of social closure in the modern age.

There is a tension between occupational closure and the values of social liberalism that promote fair access and equality of opportunity. In the cases of the professions, the justification of occupational closure involves achieving a balance between competing moral principles: on the one hand, for buildings to be safe, for accounts to be trustworthy, for all to have fair and equal representation under the law, for healthcare to be safe, and so on, it is necessary to restrict who can practice engineering, accountancy, law and medicine to those who have met a certain standard of qualification and committed to accepted standards of conduct. On the other hand, while entry into these professions may be formally open and non-discriminatory, entrants often need considerable capital—social capital in the form of social networks giving access to opportunities for apprenticeships and experience, cultural capital in the form of the ability to ‘fit in,’ be ‘taken seriously’ and generally navigate their way within professional communities, and financial capital in the form of resources to support what are often extended periods of training or low pay at the start of careers—and so the institutional structure of the professions tends to reinforce existing social inequalities. It is not by accident that the traditional British middle classes are commonly referred to as ‘the professional class’. Professional bodies, such as the British Medical Association, the Law Society, the Association of Chartered Certified Accountants, and so on, thus understand their mission as having two sides: one that focuses on ensuring standards of competence and conduct are met, and the other intervening to improve and support equitable access to the profession, through special initiatives, financial support, community engagement, communication campaigns, and research. In short, professional bodies’ interest in equality, diversity and inclusion should not be seen as a separate side project, but as a core project that ensures the social legitimacy of occupational closure, by ensuring and demonstrating that the profession is a vehicle for, and not an obstacle to, fair access.

The success of the project to professionalise cyber security in the UK will depend on the ongoing legitimacy of bodies such as the UKCSC. Cyber security is a young profession, and legitimacy cannot rest on a received understanding of its members as an elite group (as is arguably the case with doctors or lawyers). It is crucial to show how diversity will be understood and managed in the years ahead, so that achievement can be celebrated, and so that the contribution of diversity to the delivery of cyber security outcomes is part of the vision for the sector. This will require that professional bodies are attentive to, and responsive to, challenges of diversity.

3.3 The value of diversity

While considerations of social closure are sufficient to illustrate why diversity is a core priority for the legitimacy of the UKCSC, an even stronger argument can be constructed based on research into the value of diversity. Sociologists have argued that organisational or institutional diversity has value as a resource during times of economic transition (Grabher and Stark 1997). Scholars in safety studies have argued that overly homogeneous patterns of reasoning and valuation

can ‘incubate disaster’, a central problem for organisations that operate in safety critical contexts (Dekker and Pruchnicki 2014; Vaughan 1996). In Karl Weick’s classic studies of high reliability organisations, diversity can be understood as a source of ‘requisite variety’ that enables organisations to avoid accidents.

Whether team members differ in occupational specialities, past experience, gender, conceptual skills, or personality may be less crucial than the fact that they do differ and look for different things when they size up a problem. If people look for different things, when their observations are pooled they collectively see more than any one of them alone would see. However, as team members become more alike, their pooled observations cannot be distinguished from their individual observations, which means collectively they know little more about a problem than they know individually (Weick 1987).

Several studies have indicated that hiring people from diverse backgrounds provides value in broadening organisational repertoires of thought and practice, and enhancing the potential to innovate (Herring 2009). Researchers have also argued that homogeneity can be a source of problems, with evidence, for instance, that homogeneity leads to bubbles and inefficient markets (Levine et al. 2014). The 2020-2021 NCSC/KPMG Decrypting Diversity reports explicitly reference this line of thinking, arguing that diversity in the workplace brings “benefits including better financial performance, increased creativity and innovation, greater employee satisfaction, lower absenteeism and stronger talent retention” (NCSC 2020). In other words, diversity is valuable not just because inequality and discrimination are unacceptable, but also because it is a driver of innovation and ultimately a more secure society³.

3.4 The status quo

The ‘Decrypting Diversity’ reports painted a mixed picture of diversity in the cyber security profession. On the one hand, the proportion of the cyber profession that identifies as coming from a BAME (Black and Minority Ethnic) background, and the proportion that identifies as Lesbian, Gay and Bisexual (LGB), were broadly in line with national figures. Data on Trans people was insufficient to make reliable claims, and more research is needed in areas such as this. Gender is well-recognised to be an issue for cyber security, with survey respondents identifying as male outnumbering those identifying as female 2:1 in the Decrypting Diversity studies. Here, the cyber security profession needs to engage with broader efforts, as the same gender imbalance is found across the technology workforce. Socio-economic backgrounds were also skewed, for instance with those who had attended private schools over-represented. Furthermore, even in areas such as ethnicity or sexuality, where the profession was broadly in line with the general population, survey respondents noted that people from BAME backgrounds or who identified as LGB were more likely to have faced discrimination in the workplace.

The Decrypting Diversity report focuses on the experiences of people within the cyber security profession, and although it was intended to be part of a series

³Note, however, that critical voices

of annual studies, only two were carried out. Some of this research was folded into the ‘Cyber Security Skills in the UK Labour Market Survey’⁴, and like Decrypting Diversity, these ongoing surveys focus on individual diversity, and do not address connections with diversity of expertise, either at the personal level or at the level of the key problems that the cyber security profession is expected to address in work contexts. They do, however, point to the need for further research. The 2025 Cyber Security Skills in the UK Labour Market Survey pointed, for instance, to concerning indications that ethnic diversity has been trending downwards within senior cyber security roles, from 15% in 2021, to 8% in 2025.

Diversity of expertise is less well studied than individual diversity in cyber security, though recent work building on the CyBOK framework is aiming to address this. Developing a ‘knowledge profiling’ approach, Nautiyal and Rashid propose a tool for organisations to assess their internal expertise, key certifications, and expertise brought by partner organisations (Nautiyal and Rashid 2024). Other recent work uses CyBOK to analyse the ‘cyber skills gap’ through the analysis of job adverts (Attwood and Williams 2023). The Cyber Expertise Diversity Survey was developed in this vein. Like these studies, we focus on the methodological possibilities, examining how the diversity of expertise may be made visible and actionable. Our approach aims to make connections between diverse profiles of expertise among professionals, professional frameworks, and the alignment of both of these with problem scenarios.

3.5 Data analysis

3.5.1 Alignment of respondents with the specialisms

The visualisation below shows an overall view of how our respondents aligned with the UKCSC specialisms. The variance was relatively modest, with the highest average alignment roughly double that of the lowest average alignment. Looking at the individual boxplots, we note that our sample spanned a wide range, including respondents that scored very highly and very low in each area.

Of particular interest for our analysis is breaking down the overall picture according to key attributes.

3.5.2 Alignment based on key attributes

3.5.2.1 Gender

When we break out alignment by gender we notice that people identifying as men were on average better aligned than those identifying as women or non-binary in all areas apart from *Cyber security audit and assurance*, *Cyber security governance and risk management* and *Data protection and privacy*. We have ignored ‘prefer not to say’ in this case.

⁴<https://www.gov.uk/government/publications/cyber-security-skills-in-the-uk-labour-market-2025/cyber-security-skills-in-the-uk-labour-market-2025#diversity-in-cyber-security-1>

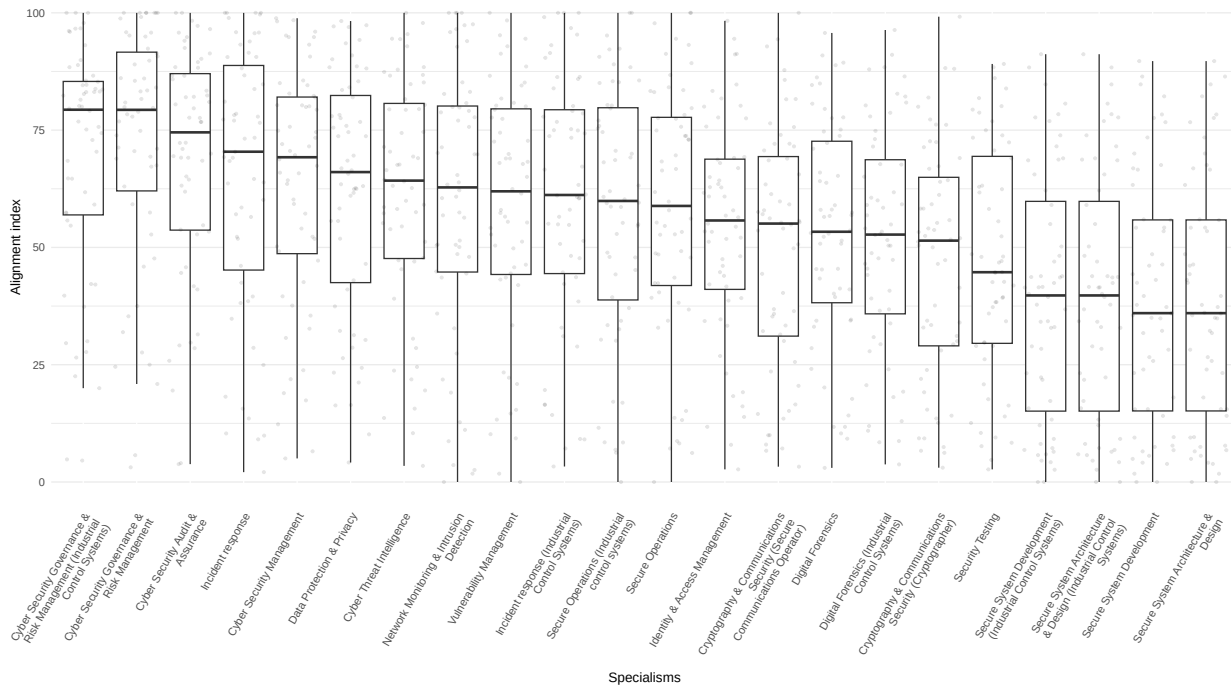


Figure 3.1: Personal alignment to specialism

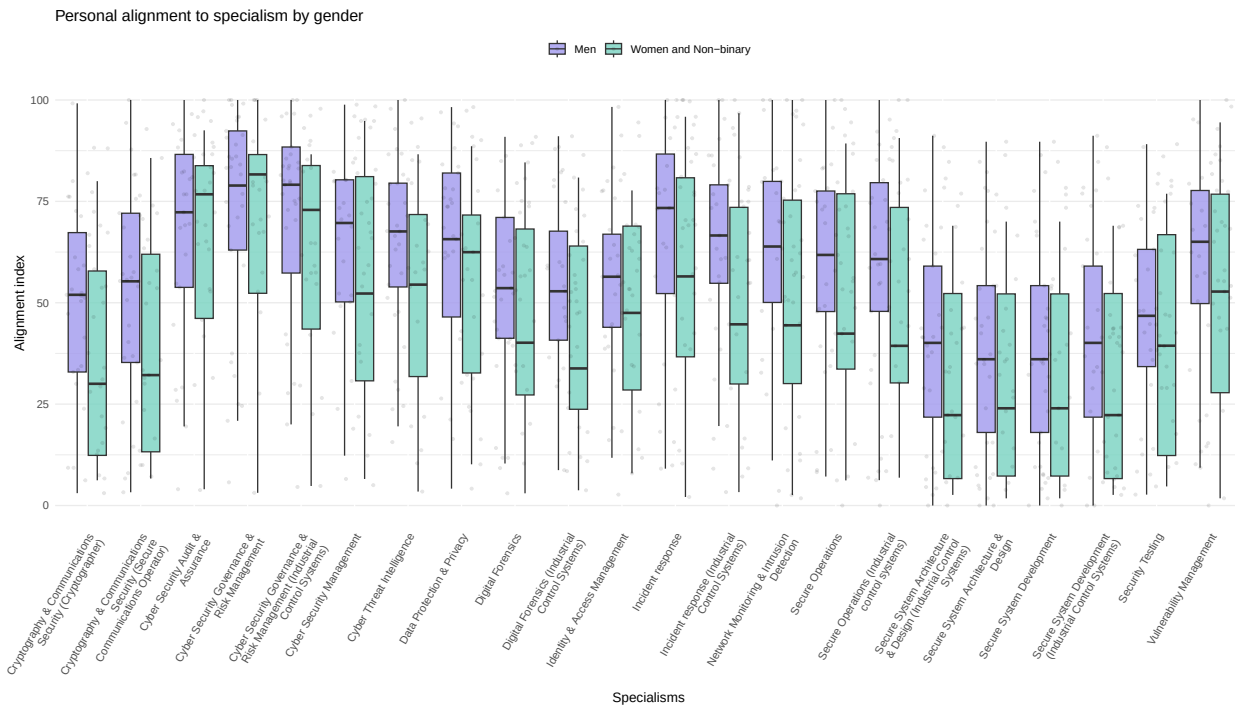


Figure 3.2: Personal alignment to specialism by gender

3.5.2.2 Age

The same view but broken down by age shows that, interestingly, in our sample younger respondents were not less well aligned than older respondents. This could reflect the fact that younger cyber security professionals may have studied degrees relating to cyber security that did not exist when older respondents were students (degrees that have syllabuses that aim at comprehensive coverage). Understanding this view from a larger dataset would be illuminating about how the profile of the profession is changing.

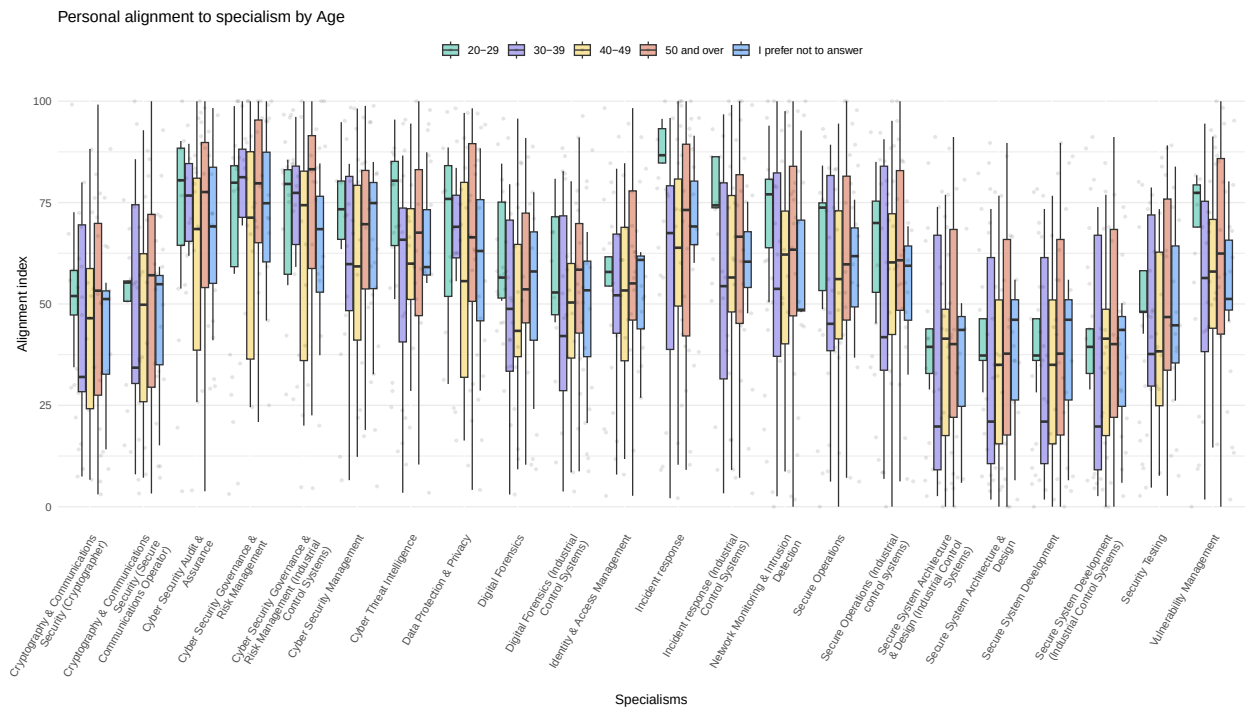


Figure 3.3: Personal alignment to specialism by age

3.5.2.3 Sexual orientation

In the visualisation below, respondents were asked ‘Do you identify as lesbian, gay or bisexual?’.

Our study shows encouraging indicative results that sexual orientation does not significantly map on to differences in alignment with specialisms.

3.5.2.4 Ethnicity

Due to having a low number of respondents in some categories, we’ve grouped data into three main categories: white, Asian and ‘all other ethnicities’.

It is potentially concerning to observe that respondents identifying as white were, on average, most well aligned with every single specialism. This observation alone could justify further research.

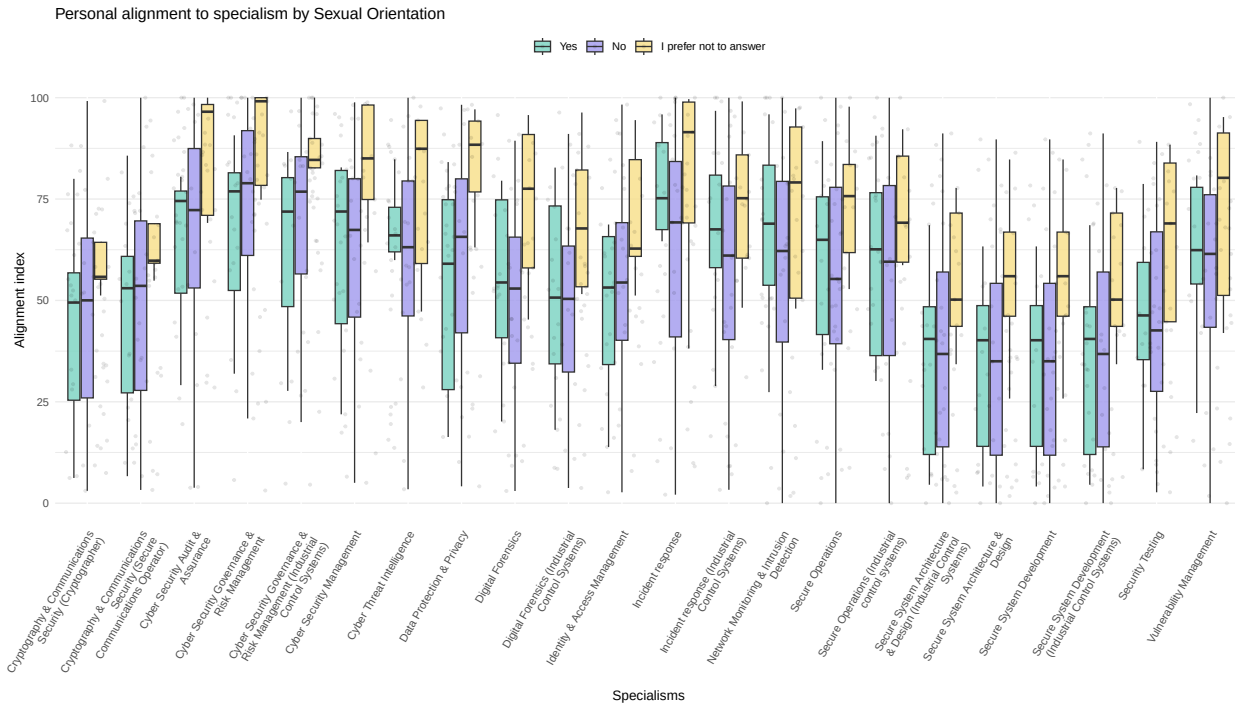


Figure 3.4: Personal alignment to specialism by sexual orientation

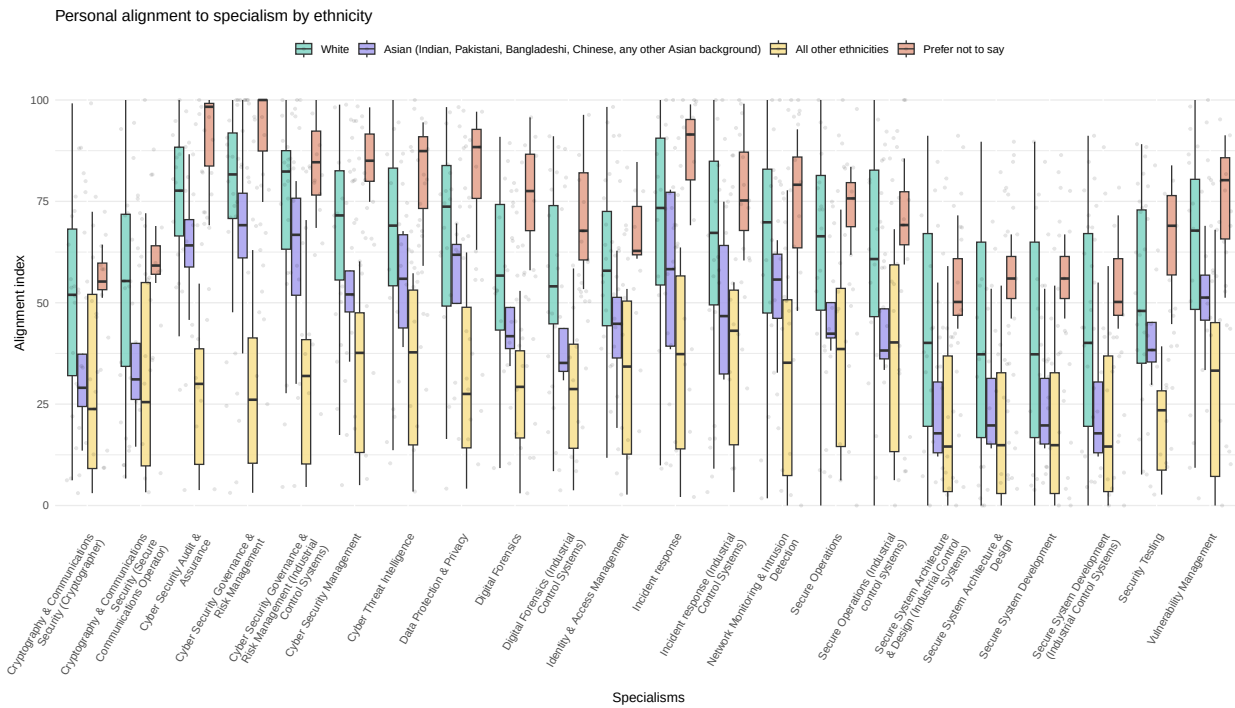


Figure 3.5: Personal alignment to specialism by ethnicity

3.5.2.5 Further analysis

The suggestive patterns we observe, particularly in relation to gender and ethnicity, warrant further investigation in order to investigate the explanation. Candidate factors include:

1. Sampling bias: Our sample may not be representative and these differences may be less apparent in a larger sample
2. Methodological factors, for instance if white and/or male respondents tend to rate their own expertise more highly than others, all else being equal.
3. Biased specialisms: The specialisms may be defined in ways that favour expertise belonging to certain groups over others
4. Structural inequalities reflected in skewed distributions of expertise across different ethnic groups

More research is needed to disambiguate these factors, and to build a dataset that would allow us to control for factors such as years of experience. In our recommendations we call for further Decrypting Diversity studies that could address this gap. In parallel with this, any specialisms that are likely to help to rebalance professional recognition in these direction should be developed. Two examples are security human factors and security awareness. Understanding what specialisms like these could look like could help future analysis to investigate the implications for inclusivity of possible future adjustments to the Cyber Careers Framework.

3.5.3 Respondents' views on Chartered status

Respondents were asked for general comments on the professionalisation of cyber security, and comments that referred directly to the Chartered status were evenly split: 5 responses were generally positive, and 5 were either negative or mixed. Starting with positive comments, respondents mentioned the same kinds of drivers around recognition and careers that have featured in the policy case for professionalisation.

"I believe that this needs to become chartered, without losing the wide range of expertise that we have in the current cohort"

"Chartered is a good step forward for the cybersecurity profession. Especially for areas where high trust is needed such as pentesting, I would like to see this done incorporating on the job expertise not only in generic cybersecurity such as with CISSP but in a certain domain or job spec."

"chartered status could be valuable if properly regulated"

"The introduction of a chartered status has been widely welcomed within the industry - I think for someone like myself who is just starting out, it

helps to define a clear pathway for necessary and relevant qualifications, and aids in providing a long-term goal / objective for my career.”

“Professionalisation of the cybersecurity industry (eg through chartered status) is welcome and overdue. I think we will continue to see promotion of CISOs to exec committees, perhaps alongside consolidation of infosec roles with wider physical security and general risk management roles. ‘Cyber’ is no longer something bolted onto the side of an organisation but is integral to most systems and processes in most organisations. Increasingly, business leaders and managers need their own skills and expertise to understand and manage cyber risks. Cybersecurity professionals need to become more versed in business risk management and develop common nomenclature and terminology to communicate directly with c-suites and boards.

I expect the uber-techie hooded hacker stereotype to become less prevalent as the field continues to mature and to appeal to a broader range of candidates.”

On the other hand, other participants offered more critical perspectives.

“I am chartered as are several of my colleagues with whom I converse regularly. There is currently absolutely no business or professional benefit to obtaining / holding any UKCSC certification.

Government procurement does not recognise expertise. Nearly all Government contracts value low cost over technical competence in almost every case and is rapidly approaching a ‘race to the bottom’ in cost terms. The Private Sector in general doesn’t care about the qualifications / certifications of their contractors.”

“The profession does not recognise or value expertise - there is an over reliance on ‘certification’ and the associated dogma that comes with it. Personnel will be hired based on having done a week long course rather than having demonstrated experience and expertise. This leads to very shallow group think and wrote responses rather than having underlying first principles knowledge and the ability to use those to make decisions and responses.”

“Certs and industry standards are a nice way to ensure a known baseline of competency, but could risk more gatekeeping in an industry already notoriously difficult to break into for new entrants”

“For me, the concern is that as a country we do not value chartered status and I don’t know how we influence that view”

“The skill of security isn’t in the expertise itself, it is being able to join the dots and think critically about what is being asked of you, dealing with peoples assumptions and also dealing with many new people coming into an industry who think they have the answer. When really cyber security has complexities and interconnections that would be very hard to create a single programme to address. If we look to silo the approach with ‘chartered’ status i would worry it does not really appreciate the multifaceted roles. We already have the issue that security only really puts value, training and gives space to the technical and those who have risk expertise. What about all the people who run between those spaces and don’t fit into the box.”

The issues raised here are wide-ranging. The potential disconnect between policy priorities and ‘on the ground’ perceptions of the value of recognition should be of concern to UKCSC and wider government policymakers. If people doubt whether professional statuses really map onto the realities of expertise and how it is valued, then they may be less willing to seek accreditation or value it in others. There is a cost to sustaining the UKCSC that will be paid by professionals, their employers, and ultimately their employers’ customers. In addition to being confident that professionalisation is, and will, deliver value for society (through reduction of risks and harms), it is necessary for the UKCSC and government policymakers to find ways to persuade professionals that they should share this confidence. This is one of the central challenges for UKCSC and its stakeholders in central government. Making adoption mandatory in certain sectors is likely to reinforce perceptions that accreditation is a cost and burden on individuals and organisations, but if a more organic process of adoption is slow, this may reinforce perceptions that the payoffs to individuals and organisations are likely to arrive far in the future. One strategy would be to focus on the value of accreditation for early career professionals (who may retain and uplift their accreditation as their careers progress).

Another strategy is indicated by figures in the recent 2025 Cyber Security Skills in the UK Labour Market survey⁵. Businesses were asked which of the Cyber Careers Framework specialisms they were finding hard to recruit for and 16% reported that they were finding it hard to fill vacancies in an area not covered by the Framework specialisms, a response that ranked equal fourth, and only behind *cyber security audit and assurance*, *cyber security governance and risk management*, and *cyber security management*. The UKCSC should ensure that cyber security professionals in all areas can see their expertise recognised in the Cyber Careers Framework. While further research is needed to be confident about which specialisms might fill this gap, security human factors and security awareness are clear candidate areas, and it may be advisable to start work developing such candidate specialisms.

⁵<https://www.gov.uk/government/publications/cyber-security-skills-in-the-uk-labour-market-2025/cyber-security-skills-in-the-uk-labour-market-2025#the-workforce-gap>

3.6 Recommendations

Specialist surveys on diversity in cyber security (Decrypting Diversity) should be revived to ensure good quality data is available. They should be extended to gather more detailed data on respondents' expertise, for example using the CyBOK framework. This would enable evidence to be gathered about correlations between diversity characteristics and the alignment of professionals with the UK Cyber Security Council Cyber Careers Framework specialisms. Regular surveys would enable the analysis of how this alignment is changing over time.

The UK Cyber Security Council should commence work on new candidate specialisms to address the current lack of coverage of security human factors and security awareness. The latter could be based on the European Cybersecurity Skills Framework 'Cybersecurity educator' role.

Chapter 4

Classifications and Diversity of Expertise

4.1 Thinking through classifications

‘To classify is human’ (Bowker and Star 1999). Standardisation smoothes out frictions: if everyone is using the same gauge of wire, the same protocols for communication, the same terminology to refer to the same thing, then the conditions are in place for collaboration and integration, and for a low risk of unforeseen complications. A classification system, can be defined, according to Bowker and Star, as ‘a set of boxes (metaphorical or literal) into which things can be put to then do some kind of work’ (Bowker and Star 1999). Devising a classification of cyber security expertise has been one of the earliest projects for the cyber security profession. The CyBOK project did this, via a large mapping exercise that ingested a wide body of data, and input from practitioners, policymakers and academics, to generate a system of 19 ‘Knowledge Areas’, published in 2019, rising to 21 in the 2021 revision (Martin et al. 2021).

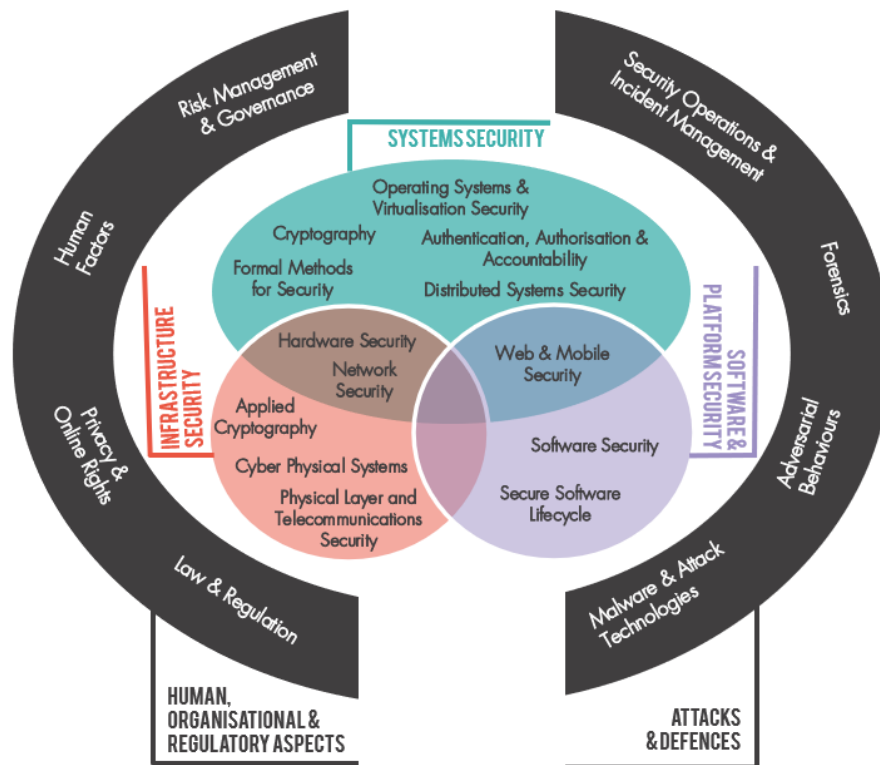


Figure 4.1: Graphic showing the CyBOK KAs (source)

CyBOK enables work to happen that would otherwise be very difficult. It provides, for instance, a framework for assessing and communicating the content of cyber security degree programmes, or for mapping out the research profiles of cyber security scholars within a university. It provides a common terminology for making comparisons and building up composite profiles.

The UKCSC created its own classification system of 15 specialisms. Where CyBOK focuses on ‘knowledge areas’, the UKCSC specialisms are intended to map onto types of cyber security job role. Each knowledge area may be relevant to many specialisms, and each specialism may require several knowledge areas. As part of its cyber careers framework, the UKCSC provided a mapping of CyBOK to the specialisms, using four levels of relevance (‘core’, ‘related’, ‘wider’ and non-relevant). Security Operations & Incident Management is the most well represented knowledge area within the UKCSC Careers Framework, core to 6 specialisms, followed by Law & Regulation, which is core to 4 specialisms. Three knowledge areas are not ‘core’ for any specialisms: Human Factors, Operating Systems and Virtualisation Security, and Physical Layer and Telecoms Security¹. Quite what, if anything, one should read into this is unclear: are these areas that are deemed less ‘in demand’ by industry? Or are they considered too specialist

¹Formal Methods and Applied Cryptography are not included due to being added in the 2021 CyBOK 1.1 update.

for the kind of typology that the Careers Framework is bringing together?

All classification systems have downsides. One downside is the propensity to create path dependencies. Classification systems are a kind of *knowledge infrastructure*, and like all infrastructure, once other systems come to depend on them, the costs of changing things later on can be high, leading to slow, incremental change and incentives against radical redesign. We encountered this challenge in our research. Our effort to expand the scope of our analysis by adding 5 new categories alongside the standard CyBOK KAs created challenges: because only the original KAs are mapped to the UKCSC specialisms, we could not incorporate new categories in our alignment calculations. Established classifications in this way condition which questions are easier and which are harder to answer.

Another downside is the fact that in prioritising some differences as the basis of their scheme, classification systems make other differences less visible. This can lead to ‘blindspots’ (to take the metaphor seriously, a blindspot is more than a gap in the visual field, it is a gap that cannot easily be perceived as a gap²). CyBOK was developed through a collaborative process that aimed to capture established knowledge areas in cyber security. While this process recognised the relevance of the social sciences to cyber security, much of the diversity of social scientific knowledge was bundled into the ‘human factors’ category. The differences in explanatory approaches across psychology, sociology, economics, organisation studies, design and anthropology therefore become harder to see. Social scientific expertise becomes invisible in the UKCSC Careers Framework, since human factors here are treated as ‘soft skills’. What was initially marginal can thus become increasingly invisible as classifications compound.

4.2 Data analysis

4.2.1 Personal expertise within and beyond CyBOK

When we asked respondents to rate their expertise and the expertise required for problem scenarios, we gave them 5 additional categories beyond those that are included in CyBOK 1.1 (Martin et al. 2021). We developed these based on discussions during the survey design process with stakeholders from government and industry, and we selected 5 to address:

- Areas that are growing in prominence, since CyBOK was first devised, and likely to continue growing. We thus included *Data Science/AI for cyber security*. AI has been considered through a CyBOK ‘topic guide’ and ‘knowledge guide’, but as yet does not have the status of a knowledge area.
- An area that is well established but happened to be left out from CyBOK. *Cloud Security* was excluded from consideration within the Network Security knowledge area. Related considerations are included within Operating System and Virtualisation Security, but the lack of prominence within CyBOK is notably at odds with the importance of cloud security expertise in industry.

²Try <https://www.scientificamerican.com/article/find-your-blind-spot/>

- Creating more space for sociotechnical cyber security expertise. CyBOK’s Human Factors knowledge area includes considerations of ‘positive security’, but ‘human factors’ is an inherently technology centric frame, referring to the ‘fit’ between the human and the machine (Meister 2018). It is conceptually a poor home for expertise based in the recognition that security may be, primarily, a matter of value and values. Security culture and security awareness, other areas of expertise that draw on ideas and methodologies from the humanities and social sciences, straddle Human Factors and Risk Management and Governance knowledge areas. To explore sociotechnical dimensions more fully, we experimented with the inclusion of:
 - *Social organisational, strategy and process*
 - *Security culture, narrative and communication*
 - *Political aspects of cyber security*

The addition of these five categories is not intended to ‘complete’ CyBOK, but to explore what we miss from the analysis when we rely on standardised ways of dividing up the world. In Figure 2.4 and Figure 2.5, *Data Science and AI for Cyber Security* was relatively low in prominence within respondents’ answers for personal and scenario combs. Understanding how areas like this may be gaining in prominence, and when they have crossed an appropriate threshold for consideration within the Careers Framework, will require that data collection steps outside of the standard categories. The other four areas, on the other hand, were widely recognised as elements of respondents’ expertise. However, while many respondents considered themselves relatively expert in *Political Aspects of Cyber Security* it is unlikely that this expertise is based in substantial understanding of political theory or political philosophy. It is a further challenge for the recognition of sociotechnical expertise in cyber security that such expertise is loosely defined.

The figure below shows on the x axis each respondent’s maximum alignment score, plotted against (on the y axis) the fraction of their comb that is covered by CyBOK. There is no significant correlation gives us confidence that the inclusion of our additional categories was not significantly skewing the personal alignment metric results.

The same cannot be said for the scenario alignment calculations. In some cases, the 5 additional categories featured prominently in the expertise required, and in these cases it is hard to say whether the UKCSC specialisms are aligned. We comment on this below.

4.2.2 Participants’ comments on classifications

The challenges of recognising and using classifications were visible to survey participants, and several comments questioned whether the categories being used were appropriate:

“The categories aren’t what I was expecting. For example, why would web be in the same category as mobile?”

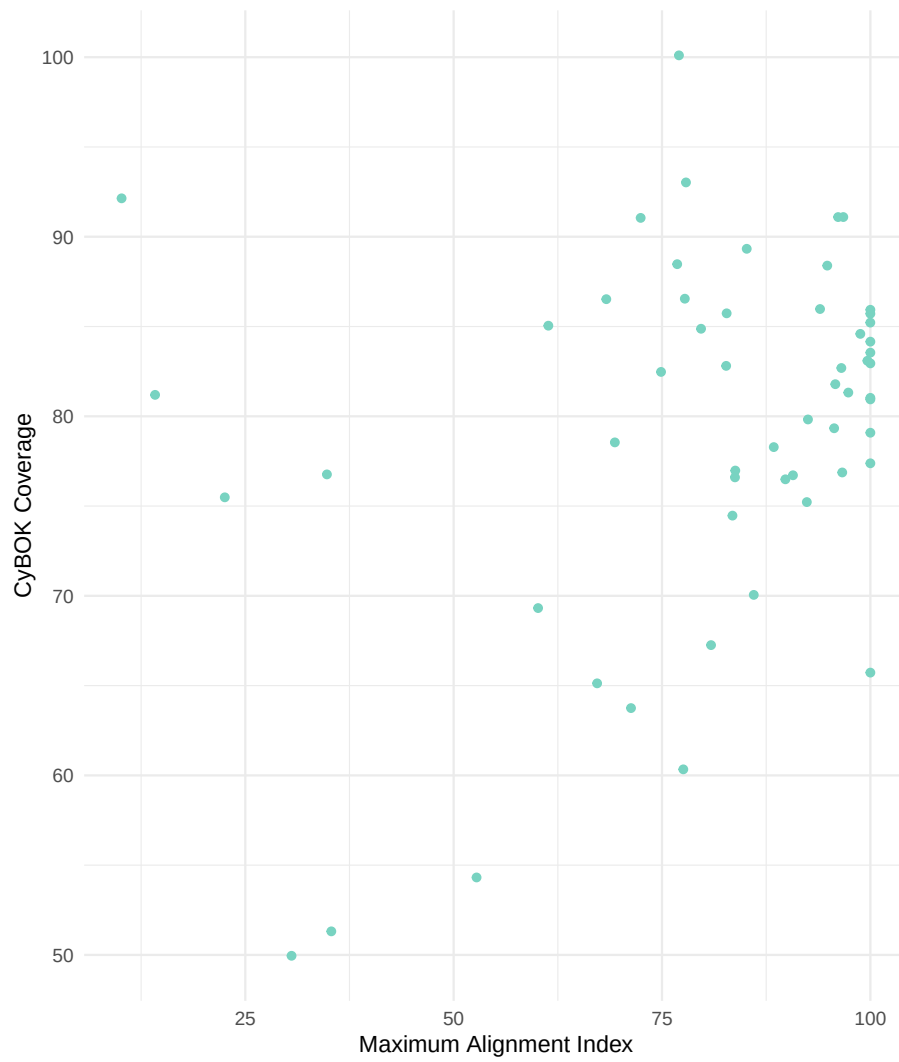


Figure 4.2: Percentage of expertise comb covered by CyBOK, versus maximum specialism alignment

“It felt like I was looking at one of a number of potential ‘combs’, i.e. I have skills and knowledge that I have used throughout my career that is not represented in the CyBOK.”

One would always expect a certain amount of friction when presenting people with classifications that reduce complexity into standard categories. However, one should also be attentive to the classification’s alignment with the field. Further research into expertise diversity in cyber security could generate input into CyBOK to help the framework address gaps and to adapt to the changing landscape of cyber security expertise and technology.

4.2.3 The value of social and human expertise

“[The profession is] Not diverse enough. Cognitively. Part of this is a function of who naturally gravitates towards the area. Some of it is nurtured by “professionalisation”, and coteries of certification cabals. Institutionalises a mind set. Exact opposite of what we need. Best people I ever recruited had humanities backgrounds (archeology or Anglo Saxon English); because they had critical thinking, communication skills and a desire to learn.”

Some degree of familiarity in human communication, behaviour, critical analysis and social explanation is picked up ‘on the job’, but having good ‘soft skills’ is entirely different from being competent in the sciences of communication, behaviour, politics and society. We do not confuse being a mathematician with general numeracy: when you train as a mathematician you do not simply become very numerate (indeed many fields of mathematics are not about numbers at all). When you train as a mathematician you learn a family of techniques and a body of knowledge. Clarifying this distinction and why it matters for cyber security should be a priority for sociotechnical security research.

“I think the cybersecurity profession focuses too much on technical security skills and not enough on transferable soft skills. We would not have such a diversity and recruitment problem in security, if we could embrace those skills like being able to manage a project, communicate with a board or problem solve. These are experiences gained in a wide range of job roles. We should value that expertise as much as the technical security skills, as without the ability to influence, communicate, change something or solve a problem, how can we drive any improvement.”

“we have an image problem in cybersecurity - we think the cool image is of a hoody wearing hacker in front of a keyboard / laptop, whether they are good guys or bad guys. The problem is this means we do not encourage the broader roles of cyber that are so not about the technology or the pen testing or hacking skills. My concern with the chartered status efforts the easy route will be to focus on chartering the technology centric aspects and roles and thus alienate

those sitting on the outskirts of the tech often driving forward change and leading major cyber security efforts.”

Respondents’ concerns with the overplayed ‘technology centric’ aspects of cyber security raises a broader issue with the technology and EDI agenda. The goal of increasing the representation of women in technology jobs has motivated campaigns to draw more women into STEM subjects in further and higher education. There is no doubt that this has been highly valuable. But there is an additional narrative that is implied here that is not necessarily helpful: that social scientific and humanities skills are not relevant to technology jobs. Changing the narrative to emphasise the contribution of the social sciences and humanities within tech and within cyber security may be needed, especially as AI technologies change skillsets. To do this effectively it will be necessary to see better representation of these forms of expertise in professional classifications.

4.3 Recommendations

Future iterations of CyBOK should be informed by research into expertise diversity, to ensure that the value of the humanities and social sciences within cyber security is better reflected. Humanities and social science researchers in the cyber security field should engage more closely with the CyBOK community to support this.

Chapter 5

Interdisciplinarity

5.1 Interdisciplinary cyber security

In this section we were interested in how one might analyse the interdisciplinary of cyber security practice. Understanding the value of diversity of expertise in cyber security requires a view of the problems that arise in cyber security, and the kinds of combinations of expertise that they require. For organisations, the utility of professional classifications depends on the extent to which it is possible to identify a need for particular kinds of expertise. Understanding the alignment of specialisms with scenarios that are representative of organisations' practical cyber security problems is one way to approach this. The Cyber Expertise Diversity Survey elicited such profiles from respondents, for a range of scenarios. Qualitative descriptions of these scenarios help us to explore the ways in which scenarios map onto specialisms, in particular:

- Which scenarios map straightforwardly onto one specialism
- Which scenarios require combinations of specialisms
- Which scenarios require significant expertise that is not accounted for by the specialisms

5.2 Data analysis

5.2.1 Respondents and scenarios

This graph plots the difference between the personal alignment metric and scenario alignment metric for each individual respondent. Each line shows the difference between these two responses. Purple shows those responses where personal skills exceed those needed for the scenario described. Teal shows those where the skills needed for the scenario exceed those of the respondent. At a very high level this gives a sense of the breadth and depth of the profession as represented in our sample.

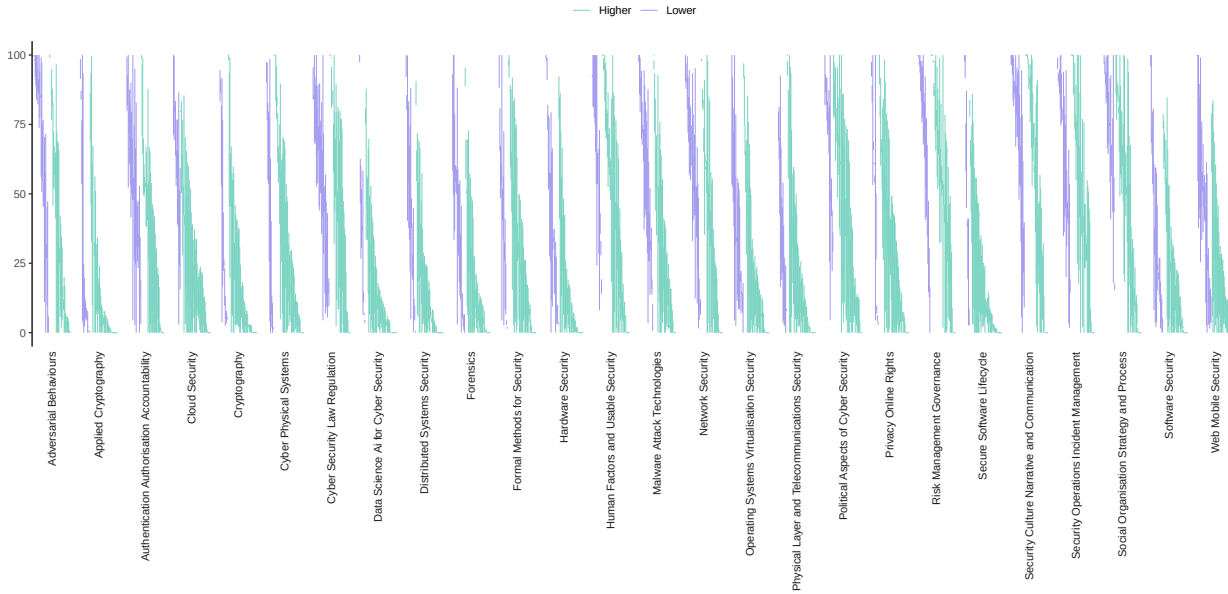


Figure 5.1: Personal vs Scenarios alignment

5.2.2 Looking at specialisms alignment with the scenarios

5.2.2.1 Alignment of permutations of multiple specialisms with problem scenarios

We calculated the alignment of the UKCSC specialisms with the scenarios our respondents told us about. We were interested in how alignment scores would increase if we allowed permutations of two or more specialisms. Figure 5.2 shows that, on average, the alignment index increases as we increase the number of permutations (this is, these scenarios are more interdisciplinary). While this could be expected, this is not the case for every scenario, as shown in Figure 5.3, which provides a more nuanced view by displaying how the alignment score changes for every scenario.

In the graphs below, a steep gradient indicates significantly better alignment could be gained through mixed teams of specialists. The alignment score for permutations of specialists is calculated by creating combined ‘combs’. We do this for all possible permutations, up to a maximum of 5, and plot how the maximum alignment for each scenario increases. The sliders for 25, 31, 44 and 53 were left blank, but we include for completeness.

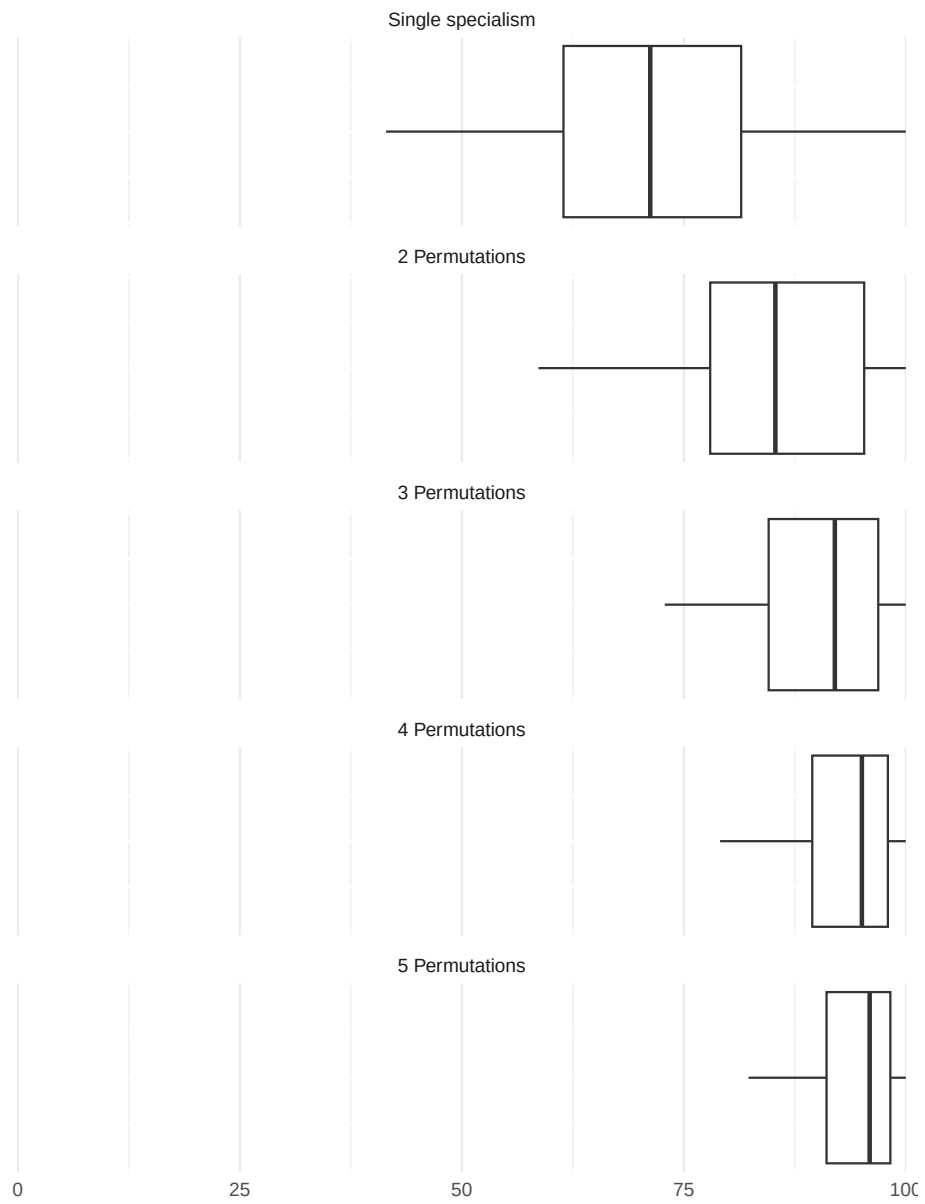


Figure 5.2: Alignment indices per number of permutations

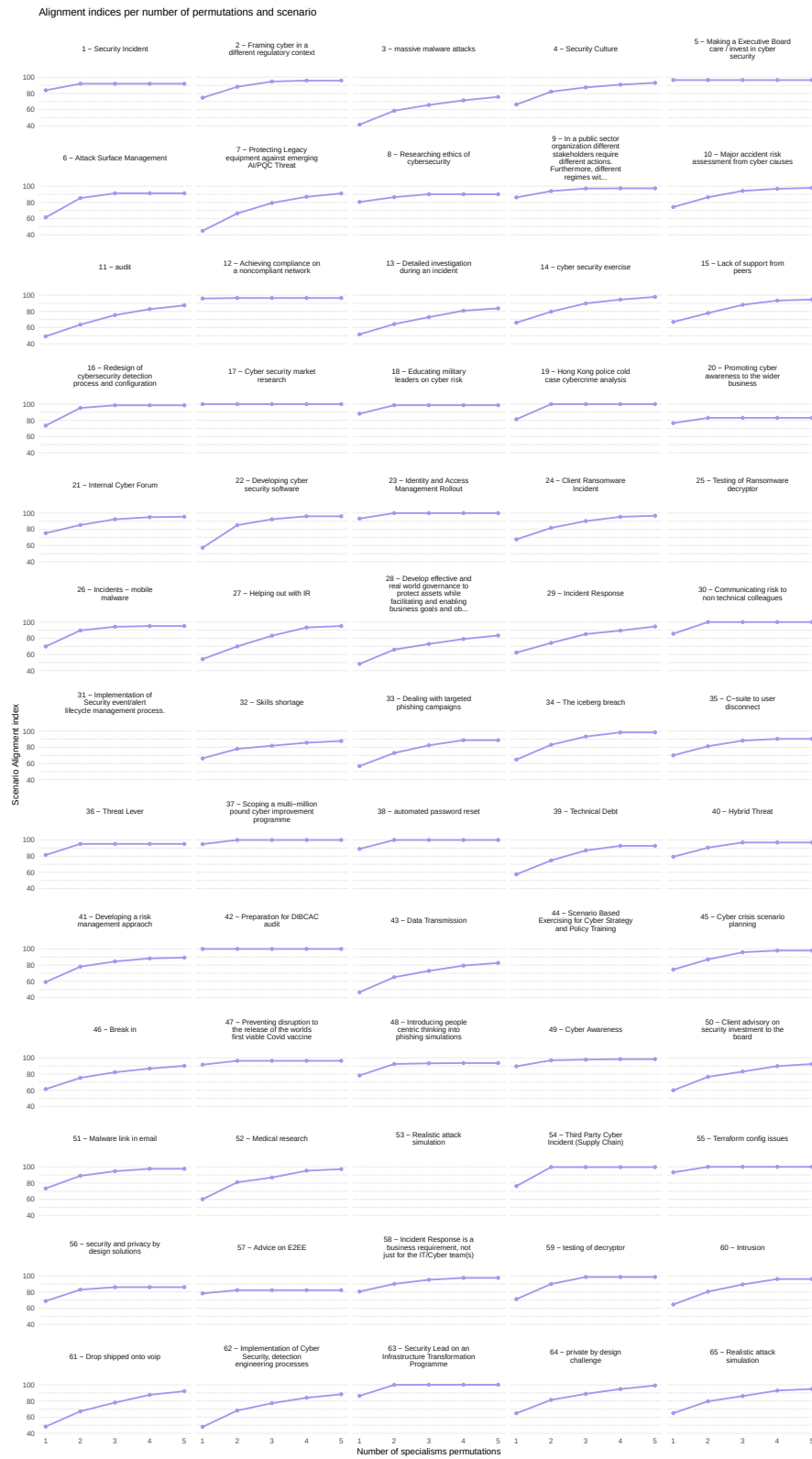
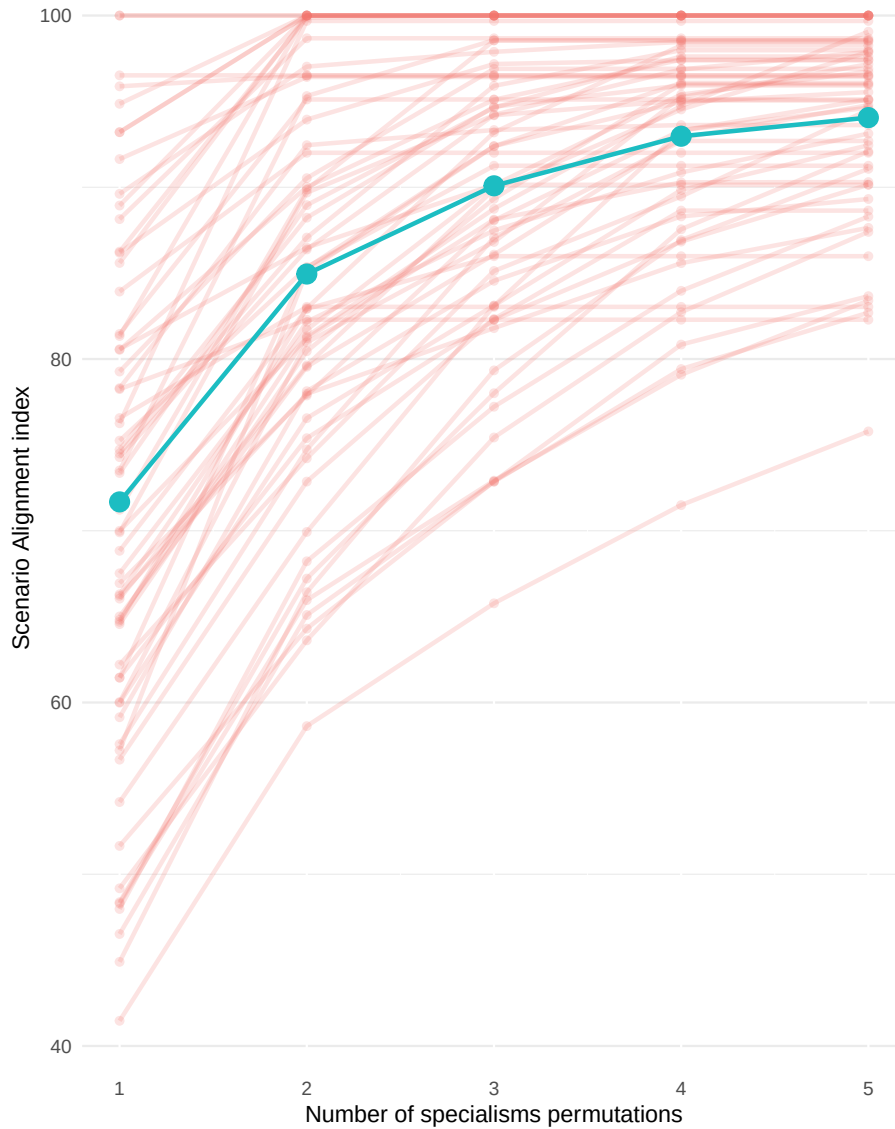


Figure 5.3: Alignment indices per number of permutations and scenario

All Alignment indices per number of permutations and scenario



5.2.3 Discussion

5.2.3.1 Lowest alignment scenarios

Scenarios that have lower overall alignment tend to be scenarios where respondents set the combs particularly ‘full’. For instance, for *Scenario 3: global mass use of malware*, the participant set all sliders to 100 (or nearly). *Scenario 13: investigation of a cyber incident*: apart from a few (data science, hardware, cyber physical and cryptography), this was deemed to require a very ‘full’ comb. Similarly, *Scenario 28: Develop effective and real world governance to protect assets while facilitating and enabling business goals and objectives* had a very

‘full’ comb.

5.2.3.2 Most interdisciplinary scenarios

Scenarios that had a steep gradient may be interpreted as particularly interdisciplinary. The adequacy of a team to tackle these scenarios improves significantly if more different specialists are added. *Scenario 7: Protecting Legacy equipment against emerging AI/PQC Threat*, for instance combined significant elements of ‘hardware security’, ‘cryptography’, ‘network security’, ‘authentication, authorisation and accountability’, and ‘distributed systems security’. Scenarios like this hit a combination of expertise that is not directly associated within existing specialisms. No UKCSC specialism, for instance, is identified with ‘authentication, authorisation and accountability’ and ‘hardware security’. This scenario is about protecting legacy equipment in a CNI context, widely understood to be a key problem area in the profession. In this case the best alignment was achieved as follows:

- 1 Specialist: Secure System Architecture & Design (Industrial Control Systems)
- 2 Specialists: Incident response (Industrial Control Systems), Cryptography & Communications Security (Cryptographer)
- 3 Specialists: Cyber Threat Intelligence, Cryptography & Communications Security (Cryptographer), Secure System Architecture & Design (Industrial Control Systems)
- 4 Specialists: Cyber Threat Intelligence, Cryptography & Communications Security (Cryptographer), Secure Operations, Secure System Architecture & Design (Industrial Control Systems)”
- 5 Specialists: Cyber Threat Intelligence, Cyber Security Management, Cryptography & Communications Security (Cryptographer), Secure Operations, Secure System Architecture & Design (Industrial Control Systems)”

Scenario 27: Incident Response was intriguing. It was also very ‘interdisciplinary’ looking at the gradient that we are using for a heuristic. But this is puzzling because there is a specialism called ‘incident response’ that one might presume ought to align well on its own. The respondent had specified the problem scenario including the requirement for very high expertise in ‘authorisation, authentication and accountability’, and quite high in ‘operating systems and virtualization security’. Neither of these are taken to be of even ‘wider’ relevance to incident response. Indeed, the incident response specialism did not feature in any of the best combinations of specialisms for this scenario. While this is highly anecdotal, it does suggest there is more to be understood, and that there could be improvements to the Cyber Careers Framework’s CyBOK mapping to ensure good alignment with practice.

Scenarios 61 and 62 were more specialist problems. Scenario 61 involved a wide spread of expertise, while scenario 62 required high expertise in a relatively high number of areas.

Got pulled from a job testing building management/HVAC systems to taking lead on assessing encrypted VoIP systems. Had to give a crash course on OT testing to more junior colleagues before switching assignments (61)

Requirement to build out a cyclical, threat detection and engineering process which required:

- Understanding of Primary Threat Adversaries and identification of their TTPs.
 - Threat Hunting capability for aforementioned TTPs (and other Indicators of Compromise).
 - Detection Engineering capability (for creation of signals, signatures and detections, specific to the aforementioned TTPs), delivering these into production.
 - Event Triage and Investigation to Incident escalation including containment, eradication and recovery.
 - Feedback mechanism of observed incidents, back into Intelligence teams to close the loop. (Detection Lifecycle Management).
- (62)

These scenarios are the more ‘naturally’ interdisciplinary cases, where problems need to be addressed that are idiosyncratic, are not neatly contained within standard specialism areas, and will either require individuals with unusual skillsets or else mixed teams.

5.2.3.3 Non-CyBOK expertise

Finally, some scenarios were specified in ways that heavily emphasised the need for non-CyBOK areas of expertise. In these cases, the alignment scores are deceptive, because we cannot calculate which specialists would contribute these skills. Interestingly, one of these cases was 58 - *Incident Response is a business requirement, not just for the IT/Cyber team(s)*.

Often people think that cyber maturity, resilience, and response is just for the cyber or IT teams to do. Promoting that it there are multiple groups that will be affected and impacted by a successful compromise and making them aware and appreciate their role to play and to consider what their individual playbook would look like is a difficult but important task to achieve. Evidence based metrics to promote the risk and then cyber resilience exercises involving key stakeholders that then promote wider engagement down the line is a necessary tool to have in your armour. (58)

Incident response is supposed to be captured within the existing specialisms, but this respondent specifies the problem in a way that emphasises expertise that is not considered within the Careers Framework. In particular, the organisational, human factors and culture aspects are deemed particularly important. Again anecdotal, examples like this are suggestive that these areas are perhaps more diverse in their needs, or in the approaches that practitioners take, than is

usually recognised. Given that several respondents voiced scepticism about the value of certifications, ensuring alignment of the specialisms with real world problems is likely to be crucial to long term success. This means examining the adequacy of the current framework, examining how problems are changing with new technology and new threats, and examining problem definition more broadly.

5.3 Recommendations

The UK Cyber Security Council's Cyber Career Framework's cross-reference to CyBOK should be refined and validated through studies of cyber security problem scenarios.

Chapter 6

Conclusions

‘More research is needed’ is never a satisfying conclusion to draw, but in a rapidly evolving sector these kinds of conclusions are crucial in ensuring that future research lines up with key questions.

- We need data that helps us to establish relationships between diversity and expertise, to influence the process of professionalisation. We have argued that this evidence base would be a key enabler for the credibility of professional categories, and ultimately for strengthening the UK’s cyber security capacity. We have argued that future Decrypting Diversity studies would be of great value to the profession especially if they incorporate attention to expertise in the way we have prototyped here.
- This data can help inform classifications such as CyBOK and the Cyber Careers Framework, ensure that gaps are filled and the profession is responsive to new areas of expertise as it adapts to changing technological and social circumstances.
- In parallel with seeking to build the evidence base, we feel it would be worthwhile for exploratory work to be undertaken to map out what new specialisms in areas such as security human factors and security awareness could look like. This would be valuable because once they are scoped out, the alignment of these specialisms with cyber security scenarios could be tested through data analysis.
- We were unable to develop a detailed analysis of the value of idiosyncratic career pathways to cyber security but this remains an issue worthy of further investigation, given that professionalisation is likely to guide the careers of future professionals along increasingly conventionalised pathways.
- We have sought to contextualise this work within bodies of literature on professionalisation, classification and diversity, which we hope will help to provide strong theoretical foundations for future analyses.

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